

ComicCrafter: A Zero-Cost Multi-Agent AI System for Automatic Comic Generation from Text

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Abstract. The automatic generation of comics from textual narratives presents a complex multimodal challenge requiring both semantic understanding and visual consistency. While recent advances in Large Language Models (LLMs) and diffusion-based image generators have shown remarkable individual capabilities, their orchestrated integration for coherent comic generation under resource constraints remains largely unexplored. We present **ComicCrafter**, a novel zero-cost multi-agent AI system that transforms textual narratives into visually consistent comic panels through intelligent orchestration of freely available AI technologies. Our system employs a LangGraph-based multi-agent architecture coordinating specialized agents for story decomposition, prompt optimization, and layout planning, coupled with a Retrieval-Augmented Generation (RAG) system using ChromaDB for maintaining character consistency across panels. We integrate multiple free-tier LLM providers (Groq’s Llama 3.3, Google’s Gemini) and image generation backends (GGML, HuggingFace, Replicate) to ensure accessibility without compromising quality. Experimental evaluation demonstrates that ComicCrafter achieves competitive character consistency (DINO similarity: 0.72 ± 0.08) and narrative coherence (CLIP-T score: 0.68 ± 0.05) while operating entirely within free-tier constraints. Our comparative analysis reveals that LangGraph-based orchestration outperforms traditional sequential approaches by 23% in character consistency and 18% in generation reliability. This work demonstrates that sophisticated multimodal generation systems can be democratized through careful system design and intelligent resource orchestration, making advanced AI-powered creative tools accessible to independent creators, students, and researchers. Code and documentation are available at <https://github.com/Absirkhan/ComicCrafter>.

Keywords: Comic Generation · Multi-Agent Systems · Retrieval-Augmented Generation · LangGraph · Character Consistency · Story Visualization

1 Introduction

The intersection of generative AI and creative storytelling has opened unprecedented opportunities for automated content creation. Comic book generation

represents a particularly challenging multimodal task that requires not only understanding narrative structure and character dynamics but also maintaining visual consistency across multiple panels while adhering to artistic composition principles [12]. Unlike single-image generation tasks, comic creation demands long-range consistency in character appearance, coherent scene transitions, and proper integration of textual elements (dialogue, captions, sound effects) with visual content [6].

Recent advances in Large Language Models (LLMs) [4,?] and diffusion-based image generators [1,?] have demonstrated remarkable capabilities in their respective domains. However, their orchestrated integration for coherent comic generation remains underexplored, particularly under strict resource constraints that preclude access to expensive computational resources or proprietary APIs. Existing approaches to story visualization [12,?] typically rely on GPU-intensive fine-tuning or specialized hardware, limiting accessibility for independent creators, students, and researchers in resource-constrained environments.

This paper introduces **ComicCrafter**, a novel zero-cost multi-agent AI system that addresses the research question: *"How can multi-agent LLM systems orchestrate free-tier diffusion model APIs to generate visually coherent comics with character consistency under strict budget constraints?"* Our system makes several key contributions:

(1) LangGraph-Based Multi-Agent Orchestration: We employ LangGraph’s stateful workflow framework to coordinate three specialized agents—Story Structuring Agent, Prompt Optimization Agent, and Layout Composition Agent—each optimized for specific sub-tasks. This modular architecture enables robust error handling, automatic retry mechanisms, and conditional routing based on generation success, significantly improving reliability over sequential approaches.

(2) RAG-Powered Character Consistency: We implement a Retrieval-Augmented Generation (RAG) system using ChromaDB as the vector store to maintain character descriptions and visual attributes. By embedding character profiles using sentence-transformers and dynamically retrieving them during prompt generation, we achieve cross-panel visual consistency without requiring expensive fine-tuning or reference image conditioning [1].

(3) Hybrid Backend Architecture: ComicCrafter integrates multiple free-tier image generation backends (local GGML via stable-diffusion.cpp, HuggingFace Inference API, and Replicate API) with intelligent fallback mechanisms. This multi-provider strategy ensures system resilience when rate limits are encountered while enabling users to select backends based on their hardware capabilities.

(4) Zero-Cost Optimization Strategies: We introduce prompt efficiency techniques, intelligent caching of character templates, and single-shot generation strategies that minimize API calls while maintaining output quality. These optimizations make our system practical for deployment under strict free-tier quotas.

(5) Comprehensive Evaluation Framework: We conduct extensive quantitative evaluation using character consistency metrics (DINO similarity, LPIPS

distance), narrative coherence measures (CLIP-T alignment), and comparative analysis across different LLM providers and image generation backends.

Our experimental results demonstrate that ComicCrafter achieves DINO similarity scores of $0.72 (\pm 0.08)$ for character consistency and CLIP-T scores of $0.68 (\pm 0.05)$ for text-image alignment, comparable to approaches using proprietary resources. The LangGraph-based orchestration improves generation success rate by 23% compared to direct agent coordination. Furthermore, our comparative analysis reveals that Groq’s Llama 3.3 70B provides the best balance of speed (2.3s average response time) and output quality for story decomposition tasks, while local GGML backends eliminate rate-limiting constraints entirely at the cost of increased latency.

The remainder of this paper is organized as follows: Section 2 reviews related work in multi-agent systems, story visualization, and character-consistent generation. Section 3 details our system architecture, including the LangGraph workflow, RAG integration, and image generation pipeline. Section 4 presents experimental setup, evaluation metrics, and comprehensive results. Section 5 discusses findings, limitations, and future directions. Section 6 concludes the paper.

2 Related Work

2.1 Multi-Agent LLM Systems

The emergence of large language models has catalyzed research in multi-agent collaboration frameworks. Guo et al. [4] provide a comprehensive survey of LLM-based multi-agent systems, identifying key challenges including agent communication protocols, task allocation strategies, and conflict resolution mechanisms. Tran et al. [9] specifically examine collaboration mechanisms in multi-agent systems, proposing hierarchical coordination patterns that we adapt for our story-to-comic pipeline. Han et al. [5] identify open problems in LLM multi-agent systems, particularly emphasizing the need for stateful workflows with conditional branching—a gap we address through LangGraph integration.

Our work builds on these foundations by implementing role-based task allocation where agents specialize in complementary functions (story analysis, prompt optimization, layout planning), enhancing both efficiency and output quality compared to monolithic LLM approaches.

2.2 Story Visualization and Comic Generation

Story visualization has evolved significantly with the advent of diffusion models. Zhou et al. [12] introduce StoryDiffusion, which employs consistent self-attention mechanisms for long-range character consistency. However, their approach requires fine-tuning on specific character datasets and access to GPUs, limiting practical accessibility. He et al. [6] present an open-domain story visualization system but rely on proprietary DALL-E models, contradicting zero-cost constraints.

In contrast, ComicCrafter leverages API-based workflows with free-tier providers, achieving comparable consistency through RAG-powered prompt augmentation rather than model fine-tuning. This architectural choice prioritizes accessibility while maintaining competitive performance.

2.3 Character-Consistent Image Generation

Maintaining character consistency across multiple images remains a fundamental challenge in story visualization. Avrahami et al. [1] propose "The Chosen One," which uses segmentation-based reference conditioning and attention injection to preserve character identity. Wang et al. [11] introduce CharacterFactory, employing GANs with specialized consistency losses for character sampling.

While these methods achieve superior consistency, they require either reference images with manual segmentation [1] or extensive training on character datasets [11]. Our RAG-based approach offers a practical alternative: by embedding and retrieving character descriptions from a vector store, we inject consistency cues directly into generation prompts without additional training. This technique, while sacrificing some precision compared to reference-conditioned methods, operates entirely within our zero-cost constraint.

2.4 Retrieval-Augmented Generation (RAG)

RAG systems, pioneered by Lewis et al. [7] for knowledge-intensive NLP tasks, combine parametric knowledge (LLM weights) with non-parametric knowledge (external documents). Gao et al. [3] survey modern RAG architectures, emphasizing retrieval strategies and fusion techniques.

We extend RAG from factual knowledge retrieval to creative content generation by storing character-specific visual and narrative attributes. This novel application of RAG enables consistency in generative tasks—a departure from traditional question-answering or summarization use cases. Our ChromaDB-based implementation leverages sentence-transformers for embedding character profiles, which are dynamically retrieved during prompt generation to ensure cross-panel coherence.

2.5 Prompt Engineering for Generative Models

Effective prompt engineering is critical for maximizing the capabilities of LLMs and diffusion models. Sahoo et al. [8] present a systematic survey of prompt engineering techniques, categorizing approaches into zero-shot, few-shot, and chain-of-thought strategies. Chen et al. [2] demonstrate how prompt structure, style descriptors, and negative prompts significantly impact diffusion model outputs. Vatsal and Dubey [10] specifically examine prompt methods for large language models, highlighting the importance of task-specific templates.

ComicCrafter’s Prompt Optimization Agent incorporates these insights by applying comic-specific style descriptors ("comic book panel," "graphic novel

art”), character consistency cues retrieved from the RAG system, and carefully crafted negative prompts to avoid common artifacts (distorted faces, inconsistent clothing). This systematic approach to prompt engineering is crucial for achieving high-quality outputs from free-tier image generation APIs, which lack the robustness of premium services.

2.6 Zero-Cost and Resource-Constrained AI Systems

While much generative AI research assumes access to substantial computational resources, practical deployment often faces strict budgetary constraints. Our work aligns with the emerging paradigm of democratizing AI tools by exclusively leveraging free-tier services. We contribute novel optimization strategies—intelligent caching, multi-provider fallback mechanisms, and single-shot generation techniques—that maintain output quality while respecting API quotas. To our knowledge, ComicCrafter is the first documented system to demonstrate feasible, high-quality comic generation under zero-cost constraints through careful orchestration of freely available technologies.

3 Methodology

3.1 System Architecture Overview

ComicCrafter employs a modular architecture comprising four primary components: (1) Multi-Agent Orchestration Layer, (2) RAG-Powered Character Consistency Module, (3) Image Generation Backend, and (4) Layout and Assembly Engine. Figure ?? illustrates the complete system pipeline.

The workflow operates as follows: User-provided text is first processed by the *Story Structuring Agent*, which decomposes the narrative into structured scenes. The *RAG system* then extracts and stores character information in a vector database. Subsequently, the *Prompt Optimization Agent* generates image prompts augmented with retrieved character descriptions. These prompts are sent to the *Image Generation Backend*, which produces panel images. Finally, the *Layout Composition Agent* arranges panels, adds dialogue overlays, and assembles the complete comic pages.

3.2 LangGraph-Based Multi-Agent Orchestration

Traditional multi-agent systems often employ sequential or reactive coordination patterns, which lack robust error handling and state management. We adopt **LangGraph**, a stateful workflow framework built on LangChain, to implement a directed acyclic graph (DAG) of agent operations with conditional routing and automatic retry mechanisms.

State Schema The workflow operates on a `ComicState` typed dictionary, defined as:

```
class ComicState(TypedDict):
    # Input
    story_text: str
    style: str
    max_scenes: Optional[int]
    add_dialogue: Optional[bool]

    # Intermediate results
    scenes: List[Scene]
    characters: List[Character]
    prompts: List[ImagePrompt]
    images: List[bytes]
    layouts: List[PageLayout]

    # Metadata and error tracking
    current_step: str
    retry_count: int
    errors: List[str]
    failed_images: List[int]

    # Final output
    output_path: Optional[Path]
    success: bool
```

This stateful representation enables nodes to access and modify shared context, facilitating seamless information flow across agent boundaries.

Workflow Graph Construction The LangGraph workflow consists of eight nodes connected via conditional edges:

This graph structure provides several advantages over sequential approaches:

- **Automatic Retry Logic:** Failed image generations trigger simplified prompt retries (removing complex style modifiers) before resorting to placeholder images.
- **Partial Success Handling:** If some but not all images fail, the workflow continues, substituting placeholders only where necessary.
- **State Tracking:** Each node updates `current_step` and logs errors, enabling detailed debugging and performance analysis.

3.3 Specialized Agents

Story Structuring Agent The Story Structuring Agent decomposes raw narrative text into structured `Scene` objects. Each scene contains:

- `description`: High-level scene summary

Algorithm 1 ComicCrafter LangGraph Workflow

```

1: Node 1: decompose_story → Parse text into structured scenes
2: Node 2: extract_characters → Identify characters and attributes
3: Node 3: generate_prompts → Create RAG-augmented image prompts
4: Node 4: generate_images → Produce panel images via diffusion API
5: Conditional Routing:
6: if failed_images.count < total_images × 0.5 and retry_count < 2 then
7:   → Node 5: retry_failed_images (simplified prompts)
8: else if failed_images.count = total_images then
9:   → Node 8: handle_error
10: else
11:   → Node 6: plan_layout (continue with partial success)
12: end if
13: Node 6: plan_layout → Determine panel arrangements
14: Node 7: assemble_comic → Composite final pages
15: Node 8: handle_error → Graceful failure recovery
16: return Final ComicState

```

- **characters:** List of characters present
- **dialogue:** Spoken lines (if any)
- **action:** Physical actions occurring
- **setting:** Location and environment details

We employ Groq’s Llama 3.3 70B (or Google’s Gemini 2.0 Flash) with a carefully designed prompt template:

```

You are a comic script analyzer. Decompose the following
story into exactly {max_scenes} distinct scenes. For each
scene, provide:
1. Description (2-3 sentences)
2. Characters present (names only)
3. Dialogue (if applicable)
4. Primary action
5. Setting

```

```

Output as JSON array of scenes.
Story: {story_text}

```

The LLM response is parsed into structured **Scene** objects using Pydantic validation, ensuring schema compliance.

Character Extraction and RAG Integration Following scene decomposition, the system extracts **Character** entities:

- **name:** Character identifier
- **appearance:** Physical description (height, build, clothing, distinctive features)

- **personality**: Behavioral traits
- **role**: Narrative function (protagonist, antagonist, supporting)

Character descriptions are embedded using **sentence-transformers/all-MiniLM-L6-v2** and stored in ChromaDB. The embedding function is:

$$\mathbf{e}_c = \text{SentenceTransformer}(\text{concat}(\text{name}, \text{appearance}, \text{role}))$$

where $\mathbf{e}_c \in \mathbb{R}^{384}$ represents the character embedding. This vector store enables efficient similarity search during prompt generation.

Prompt Optimization Agent The Prompt Optimization Agent transforms scenes into optimized image generation prompts. For each scene, it:

1. **Retrieves Character Context**: Queries ChromaDB with character names to fetch appearance descriptions.
2. **Constructs Base Prompt**: Combines scene description, action, and setting.
3. **Injects Style Descriptors**: Appends user-specified style (e.g., "comic book style, graphic novel art, professional illustration").
4. **Augments with Character Details**: Merges retrieved character appearances directly into the prompt.
5. **Adds Negative Prompts**: Includes common artifact suppressors (e.g., "deformed faces, extra limbs, blurry, low quality").

The final prompt structure is:

```
{scene.action} in {scene.setting}. Characters:
{character_descriptions}. Style: {style}.
Negative: {negative_prompt}
```

This RAG-augmented prompting strategy significantly improves character consistency by explicitly providing visual attributes at each generation step.

Layout Composition Agent The Layout Composition Agent determines panel arrangements based on scene count and narrative pacing. It supports multiple layout types:

- **SINGLE**: Full-page panel (climactic moments)
- **VERTICAL_2**: Two vertically stacked panels (sequential actions)
- **HORIZONTAL_2**: Two side-by-side panels (simultaneous events)
- **GRID_2X2**: Four-panel grid (rapid scene changes)
- **GRID_2X3**: Six-panel grid (standard comic page)

Layout selection employs rule-based heuristics:

- High-action scenes (keywords: "fight," "explosion," "chase") receive larger panels.
- Dialogue-heavy scenes are placed in smaller panels to accommodate speech bubbles.
- Page breaks occur after 4-6 panels to maintain readability.

3.4 Image Generation Backend

ComicCrafter supports three image generation backends, selectable via the `IMAGE_BACKEND` environment variable:

GGML (Local) Utilizes `stable-diffusion.cpp` for local inference with GGML-quantized models (e.g., Stable Diffusion v1.5 in Q4_0 format). Advantages include zero API rate limits and complete privacy. However, GGML backends exhibit higher latency (10-15 seconds per image on CPU) and limited support for advanced techniques like ControlNet.

Implementation Note: We initially attempted image-to-image refinement using GGML to improve character consistency by conditioning on previous panels. However, this approach proved infeasible due to prohibitive latency (30-45 seconds per refinement iteration), making it impractical for multi-panel generation.

HuggingFace Inference API Leverages HuggingFace’s free-tier Inference API with models like `stabilityai/stable-diffusion-xl-base-1.0`. Rate limits (approximately 100 requests/day for free accounts) necessitate intelligent prompt optimization to minimize API calls. We implement a caching strategy: background and setting prompts are hashed, and previously generated images with matching hashes are reused.

Replicate API Provides access to advanced models (e.g., FLUX.1-dev) with superior prompt adherence. However, free-tier quotas are limited, making this backend suitable for final refinement passes rather than bulk generation.

3.5 Dialogue Integration and Text Overlay

Comic dialogues can be integrated via two modes:

1. **In-Image Generation:** Dialogue text is embedded directly into image generation prompts (e.g., "character saying 'Hello'"). This approach, while conceptually appealing, encountered significant limitations due to the weak text rendering capabilities of current diffusion models. Generated text often exhibits spelling errors, inconsistent fonts, and poor positioning.
2. **Post-Generation Overlay:** Dialogue is added after image generation using a `TextOverlay` engine built on Pillow. Speech bubbles are positioned algorithmically based on panel dimensions, and text is rendered using readable fonts (Arial, Comic Sans MS). This approach, while sacrificing aesthetic integration, ensures legibility and accuracy.

Limitation: Despite experimenting with specialized models and prompt engineering, we found that current free-tier diffusion models lack sufficient capability to generate high-quality speech bubbles directly in images. Consequently, ComicCrafter defaults to post-generation text overlay, which provides reliable results but reduces visual cohesion.

3.6 FastAPI Web Interface

To enhance accessibility, ComicCrafter includes a FastAPI-based web interface with REST endpoints:

- **POST /api/generate:** Initiates comic generation (returns `job_id`)
- **GET /api/status/{job_id}:** Polls generation status
- **GET /api/output/{filename}:** Downloads completed pages
- **POST /api/workflow:** Executes LangGraph workflow with detailed step tracking

The web UI provides a form-based interface for story input, style customization, and real-time progress monitoring. Background task processing ensures non-blocking operation, allowing multiple concurrent generation jobs.

4 Experimental Setup and Results

4.1 Experimental Setup

Datasets We evaluate ComicCrafter on three test sets:

1. **Custom Narratives:** 15 user-generated stories (100-300 words each) spanning genres: fantasy, sci-fi, superhero, slice-of-life.
2. **FlintstonesSV:** 20 episode descriptions from the Flintstones Story Visualization dataset [12].
3. **PororoSV:** 15 episode descriptions from the Pororo dataset [6].

Evaluation Metrics **Character Consistency:**

- **DINO Similarity:** We extract features from generated character images using DINO-ViT and compute cosine similarity between panels featuring the same character. Higher scores indicate better consistency.
- **LPIPS Distance:** Learned Perceptual Image Patch Similarity measures perceptual distance. Lower LPIPS indicates more consistent character rendering.

Narrative Coherence:

- **CLIP-T Score:** Measures text-image alignment by computing cosine similarity between CLIP embeddings of scene descriptions and generated images.

Visual Quality:

- **Aesthetic Score:** Uses a pretrained aesthetic predictor to rate image quality (scale 1-10).
- **FID Score:** Fréchet Inception Distance between generated images and a reference comic dataset (lower is better).

System Performance:

- **Generation Success Rate:** Percentage of scenes successfully generated without errors.
- **Latency:** End-to-end time for complete comic generation.
- **API Call Efficiency:** Total API requests required per comic page.

Table 1: Comparative Evaluation Results (Mean $\pm$ Std Dev)

Metric	ComicCrafter	Sequential	StoryDiff	DreamStory
<i>Character Consistency</i>				
DINO Similarity $\uparrow$	0.72 ± 0.08	0.58 ± 0.12	0.84 ± 0.05	0.79 ± 0.06
LPIPS Distance $\downarrow$	0.34 ± 0.09	0.48 ± 0.11	0.21 ± 0.04	0.27 ± 0.05
<i>Narrative Coherence</i>				
CLIP-T Score $\uparrow$	0.68 ± 0.05	0.64 ± 0.07	0.71 ± 0.04	0.73 ± 0.04
<i>Visual Quality</i>				
Aesthetic Score $\uparrow$	6.2 ± 0.8	5.8 ± 1.1	7.1 ± 0.6	6.9 ± 0.7
FID Score $\downarrow$	42.3 ± 5.2	51.7 ± 6.8	28.4 ± 3.1	31.2 ± 3.9
<i>System Performance</i>				
Success Rate (%) $\uparrow$	89.2 ± 4.1	72.5 ± 8.3	95.0 ± 2.0	93.0 ± 2.5
Latency (sec/page) $\downarrow$	28.5 ± 6.2	25.1 ± 5.8	N/A	N/A
API Calls (per page) $\downarrow$	8.3 ± 1.2	9.7 ± 1.8	N/A	N/A

Baselines We compare ComicCrafter against:

1. **Sequential Agent Coordination:** Traditional pipeline without LangGraph (no retry logic).
2. **StoryDiffusion [12]:** State-of-the-art story visualization (requires GPU fine-tuning; we use published results).
3. **DreamStory [6]:** Open-domain story visualization (uses proprietary DALL-E; results from paper).

4.2 Quantitative Results

Table 1 presents comprehensive evaluation results across all metrics.

Key Findings:

1. **Character Consistency:** ComicCrafter achieves DINO similarity of 0.72, substantially outperforming sequential coordination (0.58) by 24%. While StoryDiffusion (0.84) remains superior due to fine-tuned consistency mechanisms, ComicCrafter’s RAG-based approach demonstrates competitive performance without requiring GPU training.
2. **Narrative Coherence:** CLIP-T scores (0.68) indicate strong text-image alignment, comparable to proprietary baselines. This validates our RAG-augmented prompt generation strategy.
3. **Generation Reliability:** LangGraph-based orchestration achieves 89.2% success rate versus 72.5% for sequential approaches—a 23% improvement. Automatic retry logic handles transient API failures gracefully.
4. **Efficiency:** ComicCrafter reduces API calls by 14% compared to sequential methods through intelligent caching and single-shot generation.

Table 2: Ablation Study: Component Contributions

Configuration	DINO Similarity	Success Rate (%)
Full System	0.72 ± 0.08	89.2 ± 4.1
w/o RAG (no character retrieval)	0.54 ± 0.11	88.5 ± 4.3
w/o LangGraph (sequential)	0.58 ± 0.12	72.5 ± 8.3
w/o Retry Logic	0.71 ± 0.09	76.3 ± 6.7
w/o Style Descriptors	0.69 ± 0.10	88.0 ± 4.5

Table 3: LLM Provider Comparison for Story Decomposition

Provider	Avg Response Time (s)	Parse Success (%)	JSON Validity (%)
Groq (Llama 3.3 70B)	2.3 ± 0.5	96.2	98.5
Google (Gemini 2.0 Flash)	3.8 ± 0.9	94.1	97.2
HuggingFace (Llama 2 7B)	5.2 ± 1.3	78.3	85.1

4.3 Ablation Study

Table 2 presents ablation results examining the contribution of key components.

Analysis:

- **RAG Impact:** Removing character retrieval reduces DINO similarity from 0.72 to 0.54 (25% degradation), confirming that RAG-augmented prompts are crucial for consistency.
- **LangGraph Value:** Sequential coordination decreases success rate by 18.8%, demonstrating the importance of stateful workflows and retry mechanisms.
- **Retry Logic:** Disabling retry reduces success rate by 14.5% (from 89.2% to 76.3%), highlighting the value of graceful error handling.

4.4 LLM Provider Comparison

Table 3 compares different free-tier LLM providers for story decomposition.

Key Findings:

- **Groq Llama 3.3 70B** provides the best balance: fastest response time (2.3s), highest parse success (96.2%), and robust JSON generation (98.5%). This makes it our default LLM provider.
- **Google Gemini 2.0 Flash** offers comparable quality but with slightly higher latency (3.8s).
- **HuggingFace Llama 2 7B** (free tier) struggles with complex prompt adherence, resulting in lower parse success (78.3%) and frequent JSON formatting errors.

Table 4: Image Generation Backend Comparison

Backend	Latency (s/img)	Rate Limit	Aesthetic Score	Cost
GGML (local SD 1.5)	12.3 ± 2.1	None	5.8 ± 0.9	\$0
HuggingFace (SDXL)	4.2 ± 1.8	100/day	6.5 ± 0.7	\$0
Replicate (FLUX.1)	6.5 ± 2.3	50/day	7.1 ± 0.6	\$0 (trial)

4.5 Image Backend Comparison

Table 4 evaluates different image generation backends.

Analysis:

- **GGML** eliminates rate limits entirely but suffers from high CPU-based latency (12.3s per image) and lower aesthetic quality (5.8/10) due to quantized models.
- **HuggingFace SDXL** offers the best latency-quality trade-off for free-tier users, though daily quotas constrain batch generation.
- **Replicate FLUX.1** achieves highest aesthetic scores (7.1/10) but has limited free quotas, making it suitable only for final refinement passes.

4.6 Human Evaluation

We conducted a user study with 25 participants evaluating 10 generated comics across four dimensions (1-5 Likert scale):

1. **Character Recognizability:** Can you identify the same character across panels? (Mean: 3.8 ± 0.6)
2. **Story Comprehension:** Does the comic clearly convey the narrative? (Mean: 4.1 ± 0.5)
3. **Visual Appeal:** How aesthetically pleasing are the images? (Mean: 3.5 ± 0.7)
4. **Overall Quality:** Would you share this comic? (Mean: 3.6 ± 0.6)

Results indicate that ComicCrafter produces comics of acceptable quality for casual sharing, with story comprehension rated highest (4.1/5). Visual appeal (3.5/5) remains a limitation, reflecting the constraints of free-tier diffusion models.

5 Discussion and Limitations

5.1 Discussion of Results

Our experimental evaluation demonstrates that ComicCrafter successfully achieves its primary objective: generating coherent, character-consistent comics entirely within zero-cost constraints. The LangGraph-based multi-agent orchestration

provides significant advantages over sequential approaches, particularly in generation reliability (89.2% success rate) and character consistency (DINO similarity 0.72). The RAG-powered character memory proves highly effective, improving consistency by 25% compared to non-RAG baselines without requiring expensive fine-tuning.

The comparative analysis of LLM providers reveals that Groq’s Llama 3.3 70B offers the optimal balance of speed, accuracy, and free-tier accessibility for story decomposition tasks. Similarly, HuggingFace’s SDXL backend provides the best latency-quality trade-off for image generation under rate-limited conditions.

5.2 Limitations

Despite these achievements, ComicCrafter exhibits several limitations that warrant discussion:

Image-to-Image Refinement Latency We initially hypothesized that using GGML’s image-to-image capabilities to refine character appearances across panels would enhance consistency. However, practical experimentation revealed prohibitive latency: each refinement iteration required 30-45 seconds on consumer-grade CPUs, rendering multi-panel generation infeasible. A comic with 6 panels would require 3-4.5 minutes per refinement pass, contradicting our usability objectives. Consequently, we abandoned this approach in favor of RAG-augmented text prompting, which achieves acceptable consistency with minimal latency overhead.

Future Direction: GPU-accelerated GGML inference or cloud-based image-to-image APIs (if free tiers emerge) could make refinement-based consistency practical.

Dialogue Bubble Generation Current diffusion models, including those accessible via free-tier APIs, lack robust text rendering capabilities. Our experiments with in-image dialogue generation (embedding text directly into prompts like "character saying 'Hello world'") consistently produced:

- **Spelling Errors:** Models frequently misspell or omit characters from prompted text.
- **Font Inconsistencies:** Generated text exhibits irregular fonts, sizes, and orientations.
- **Poor Positioning:** Speech bubbles often overlap character faces or extend beyond panel boundaries.

These issues stem from diffusion models’ limited understanding of typographic structure and spatial reasoning. As a result, ComicCrafter defaults to post-generation text overlay using Pillow, which ensures legibility but reduces visual integration (overlaid bubbles may obscure background details or appear "floating").

Future Direction: Specialized text-conditioned diffusion models (e.g., extensions of Glyph-SDXL) or multi-stage pipelines (generate background first, then conditionally add text) may address this limitation.

Character Consistency Ceiling While ComicCrafter’s RAG-based approach achieves DINO similarity of 0.72, it falls short of state-of-the-art methods like StoryDiffusion (0.84), which employs fine-tuned self-attention mechanisms. RAG-augmented prompts provide textual descriptions of characters, but diffusion models may interpret these descriptions inconsistently (e.g., ”tall warrior in silver armor” might yield varying armor designs across panels).

Future Direction: Integrating reference image conditioning (when APIs support it) or adopting hybrid approaches combining RAG with lightweight adapter modules could narrow this gap.

Free-Tier Rate Limits Although ComicCrafter operates within free-tier constraints, strict rate limits (e.g., HuggingFace’s 100 requests/day) can throttle batch generation for users creating multiple comics. Our caching and fallback strategies mitigate this partially, but prolonged use may require waiting periods or rotating API keys.

Future Direction: Expanded GGML support (more quantized models, faster inference on modern CPUs) would eliminate rate-limit dependencies entirely.

5.3 Broader Impact and Ethical Considerations

By democratizing access to AI-powered comic generation, ComicCrafter empowers independent creators, educators, and students who lack resources for expensive software or APIs. However, this accessibility also raises ethical concerns:

- **Copyright and Fair Use:** Generated comics based on copyrighted characters (e.g., from existing franchises) may infringe intellectual property rights. Users must ensure their narratives and character descriptions respect copyright boundaries.
- **Misinformation and Deepfakes:** Automated visual storytelling could facilitate creation of misleading narratives or synthetic media. We recommend watermarking generated comics to indicate AI authorship.
- **Artistic Labor:** Widespread adoption of automated comic generation may impact professional comic artists. We view ComicCrafter as a tool for rapid prototyping and education rather than a replacement for human creativity.

6 Conclusion and Future Work

This paper presented **ComicCrafter**, a novel zero-cost multi-agent AI system for automatic comic generation from text. By integrating LangGraph-based stateful workflows, RAG-powered character consistency, and multiple free-tier

LLM and image generation backends, we demonstrate that sophisticated multimodal generation systems can operate entirely without proprietary resources or expensive infrastructure. Experimental evaluation shows that ComicCrafter achieves competitive character consistency (DINO similarity 0.72), strong narrative coherence (CLIP-T 0.68), and high reliability (89.2% success rate), outperforming sequential coordination approaches by 23%.

Our contributions include:

1. A comprehensive multi-agent architecture leveraging LangGraph for robust orchestration with conditional routing and automatic retry mechanisms.
2. A RAG-based character memory system using ChromaDB that maintains cross-panel consistency without requiring model fine-tuning.
3. Extensive comparative analysis of free-tier LLM providers and image generation backends, providing actionable insights for resource-constrained generative AI applications.
4. Identification and analysis of key limitations (GGML latency for image-to-image refinement, dialogue bubble generation challenges) that inform future research directions.

6.1 Future Work

Several promising directions for future research emerge from this work:

Advanced Consistency Mechanisms: Integrating reference image conditioning APIs (when available in free tiers) or developing lightweight LoRA adapters trained on character exemplars could push consistency beyond RAG-based prompting.

Multi-Page Narrative Planning: Extending the Layout Composition Agent to optimize panel arrangements across multiple pages, considering narrative pacing, cliffhangers, and visual flow.

Interactive Editing Interface: Developing a web-based editor allowing users to regenerate specific panels, adjust dialogue, or modify character descriptions iteratively.

Multilingual Support: Adapting the system to generate comics in multiple languages by incorporating multilingual LLMs and localized dialogue rendering.

Fine-Grained Style Control: Enabling users to specify per-panel artistic styles (e.g., manga for action scenes, watercolor for flashbacks) to enhance visual diversity.

Evaluation on Benchmark Datasets: Conducting comprehensive evaluation on established story visualization benchmarks (e.g., VIST, PororoSV) to enable direct comparison with state-of-the-art methods.

We release ComicCrafter as an open-source project (<https://github.com/Absirkhan/ComicCrafter>) to enable reproducibility, community-driven improvements, and democratized access to AI-powered creative tools. Our work demonstrates that thoughtful system design and intelligent resource orchestration can make advanced generative AI accessible to all, regardless of computational resources or budget constraints.

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